

The Listening City System: A Smart Citizen Feedback Platform for Urban Incident Reporting

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Abstract. The Listening City System is a smart citizen feedback platform designed to support urban incident reporting and response management. The system allows citizens to submit reports about public issues such as traffic accidents, fire incidents, damaged infrastructure, environmental problems, and security concerns. Each report may include a description, GPS location, images, or videos to help authorities verify and process the issue more effectively.

The platform supports different user roles, including citizens, ward staff, and police staff. Citizens can create reports, track processing status, receive notifications, and provide feedback after the issue is handled. Ward staff and police staff can receive assigned reports, update processing results, and coordinate responses based on incident type and location. The system aims to improve communication between citizens and local authorities, reduce manual reporting procedures, and increase transparency in urban management.

Keywords: Smart City · Citizen Feedback · Urban Incident Reporting · GPS · Public Service Management

1 Introduction

Urban areas often face many public issues such as traffic accidents, fire incidents, damaged roads, flooding, waste problems, and security concerns. Traditional reporting methods usually depend on phone calls, paper forms, or direct visits to local authorities. These methods may be slow, difficult to track, and lack clear evidence.

The Listening City System is proposed as a smart platform that allows citizens to report urban incidents quickly through a web-based system. By using GPS location, images, videos, and digital status tracking, the system helps authorities receive and process reports more efficiently.

2 Problem Statement

In many cities, citizens may not know the correct department to contact when an incident occurs. Reports can be delayed, duplicated, or sent to the wrong

authority. In addition, local staff may have difficulty verifying whether a report is real or fake if there is not enough supporting evidence.

Therefore, a centralized feedback platform is needed to collect citizen reports, classify incident types, assign reports to suitable authorities, and provide transparent status updates.

3 Related Work

Existing smart city systems often focus on citizen engagement, public service management, and digital government services. Many systems allow residents to submit complaints or report public issues online. However, some systems still lack real-time location support, clear workflow management, and direct coordination between citizens and local authorities.

The Listening City System combines citizen feedback, GPS-based reporting, evidence submission, and authority workflow management to improve urban incident handling.

4 System Analysis and Design

The main actors of the system include Citizen, Ward Staff, and Police Staff.

Citizens can register, log in, submit incident reports, upload images or videos, track report status, and give feedback after the issue is resolved.

Ward Staff can receive reports related to local public issues, verify information, update processing status, and provide final results.

Police Staff can handle urgent or security-related incidents such as fire alerts, accidents, and public safety problems.

5 Proposed Solution

The proposed system provides a web-based platform where citizens can submit reports with incident type, description, GPS location, and media evidence. The system stores report information in a database and supports report assignment based on category and location.

For urgent incidents, the system can provide a fast reporting flow so that important information is sent quickly to the responsible authority. To reduce fake reports, the system may require user authentication, OTP verification, evidence upload, and report history tracking.

6 Implementation Plan

The system can be developed using a front-end and back-end architecture. The front-end provides user interfaces for citizens and staff. The back-end handles authentication, report management, assignment workflow, notification, and database operations.

The database stores user accounts, report categories, incident reports, locations, media files, status history, and feedback records.

7 Conclusion

The Listening City System aims to improve communication between citizens and local authorities by providing a digital platform for urban incident reporting. The system helps citizens report problems faster and allows authorities to manage reports more transparently and efficiently. In the future, the system can be extended with AI-based report classification, dashboard analytics, and integration with emergency services.

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